Drug Review Sentiment Analysis using NLP

Python | VADER | BERT | Healthcare NLP

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Business Problem

- Drug reviews contain valuable patient feedback.
- Can NLP identify patient sentiment?
- Can NLP uncover meaningful insights from patient drug reviews?

Dataset

- DrugLib dataset
- ~4,000+ reviews
- Features
  - Drug
  - Condition
  - Benefits
  - Side Effects
  - Comments
  - Rating

PROJECT WORKFUL

Dataset
↓
Data Cleaning
↓
Text Preprocessing
↓
VADER Sentiment Analysis
↓
BERT Sentiment Classification
↓
Sentiment Comparison
↓
Healthcare Insights

NLP Techniques Used

- *Data Cleaning*
- *Tokenization*
- *Stopword Removal*
- *Lemmatization*
- *VADER*
- *BERT*

VADER Sentiment Analysis Results

Distribution of positive, neutral, and negative sentiment across patient drug reviews using the VADER sentiment analysis model.

```
Sentiment analysis for: commentsReview
vader_label
positive    36.345581
negative    35.451837
neutral     28.202582
Name: proportion, dtype: float64

Sentiment analysis for: benefitsReview
vader_label
positive    46.772592
negative    38.828203
neutral     14.399206
Name: proportion, dtype: float64

Sentiment analysis for: sideEffectsReview
vader_label
negative    54.419067
neutral     25.521351
positive    20.059583
Name: proportion, dtype: float64
```

Key Findings

- Benefits reviews showed the highest positive sentiment (46.8%).
- Side effects reviews showed the highest negative sentiment (54.4%).
- Comments reviews were relatively balanced across all three sentiment classes.

BERT Sentiment Classification Results

Sample patient reviews classified using a BERT transformer model to identify sentiment from unstructured healthcare text

	benefitsReview	benefitsReview_raw	benefitsReview_sentiment
0	the antibiotic may have destroyed bacteria cau...	3 stars	neutral
1	lamictal stabilized my serious mood swings. on...	5 stars	positive
2	initial benefits were comparable to the brand ...	4 stars	positive
3	it controlls my mood swings. it helps me think...	5 stars	positive
4	within one week of treatment superficial acne ...	1 star	negative

Key Insights

- BERT successfully classified patient reviews into **positive**, **neutral**, and **negative** sentiment.
- Reviews with **higher ratings (4–5 stars)** were generally classified as **positive**.
- Reviews with **1-star ratings** were correctly identified as **negative**, demonstrating BERT's ability to understand contextual meaning rather than relying solely on keywords.

Business Impact

Example:

NLP enables healthcare organizations to analyze thousands of patient reviews automatically, helping identify medication experiences, patient satisfaction trends, and potential adverse effects from real-world feedback.

Future Improvements

- Fine-tune BERT on larger healthcare datasets.
- Deploy the model as a web application.
- Expand to additional real-world drug review datasets.
- Incorporate explainable AI techniques.